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Analysis of the sporozoite ELISA for estimating infection rates in Mozambican anophelines

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Abstract. Comparisons were undertaken to investigate cost-effective methods of implementing the enzyme-linked immunosorbent assay (ELISA) for sporozoite determination in anophelines when large numbers require processing. Comparisons between ELISA plate reader and visual assessments were performed with *Anopheles funestus* and *Anopheles gambiae s.l.* (Diptera: Culicidae), as were comparisons between whole-body mosquito samples, heads and thoraces, and abdomens alone. Rates obtained from pools of five or 10 mosquitoes were compared with those for individual mosquitoes, as were rates obtained using different sampling methods. A total of 41 792 *An. funestus* and 9431 *An. gambiae s.l.* collected in light traps, and 22 323 *An. funestus* and 6860 *An. gambiae s.l.* from exit collections were analysed. Visual assessments gave results similar to those of machine readings. Sporozoite rates were similar in both species, as were rates by collection method. The use of whole mosquitoes increased estimates of infection rate by 0.6%. Pool size did not affect infection rates of *An. gambiae s.l.*, but rates were higher among individually tested *An. funestus* than among those tested in pools. For large-scale surveys, the use of whole mosquitoes in pools of 10 mosquitoes, with correction for overestimation, and the noting of results according to a simple three-stage visual assessment of positivity is the most cost-effective approach and is sufficient to obtain reliable data for comparative purposes.

Key words. *Anopheles funestus*, *Anopheles gambiae*, circumsporozoite-ELISA, sporozoite, Mozambique.

Introduction

The number of infectious bites a person receives in a given time, or the entomological inoculation rate (EIR), is used in the epidemiological monitoring of malaria control efforts. Calculations of the EIR involve the determination of vector densities and mosquito infection rates. The EIR, a critical parameter in the epidemiology of malaria, is obtained by multiplying the number of mosquitoes biting an 'average' person by the proportion of these that are infected with infectious sporozoites.

In order to determine the effect of an intervention on transmission, regular surveys of the infection status of the vector(s) are required. Such monitoring efforts must be intensified as transmission is reduced. Indeed, as efforts to eliminate malaria intensify, exophagic and less efficient vectors that may only transmit malaria when they occur in large numbers may maintain a low level of transmission. This may lead to a situation in which the often endophagic primary vectors previously affected by insecticides develop resistance and populations may recover sufficiently to reinitiate high levels of malaria transmission. Large

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numbers of insects must be processed in order to verify which, if any, of these so-called secondary vectors might be responsible for maintaining transmission. Similarly, methods of malaria control that do not rely on vector control (such as vaccines) may lead to *anophelism sans malaria*, in which case the EIR even in primary vectors may reach very low levels.

The methods currently used to estimate biting rates provide relative rather than absolute estimates of numbers of bites received. This is of less importance than it might at first seem because comparative estimates are required (such as in studies that compare intervention and control settings), and it matters less which methods are used than that they are standardized across sites or times (Dye, 1992). The same is true for the estimation of sporozoite rates.

Although the dissection of salivary glands may be considered the reference standard method of determining infection rates, it has a number of disadvantages when large sample sizes must be processed. Freshly killed mosquitoes are required for dissection, the method of which requires a relatively high level of skill, and the species of sporozoite is not known (with the result that non-human infections may be misclassified). Techniques based on polymerase chain reaction (PCR) represent an alternative, but these also require relatively sophisticated equipment and personnel. The most widely used method to determine sporozoite rates is the circumsporozoite protein enzyme-linked immunosorbent assay (cs-ELISA) developed by Burkot *et al.* (1984) and Wirtz *et al.* (1987). The reliability of the ELISA to determine sporozoite rates has, however, recently been questioned by Durnez *et al*. (2011), who found a number of false positives in zoophilic mosquitoes collected from Southeast Asia. False positives have also been associated with zoophilic anophelines from Africa (Koekemoer *et al*., 2001) and bovine-fed mosquitoes (Lochouarn & Fontenille, 1999). These false positives may result from cross-reactivity with closely related parasites of veterinary importance. For the major African vectors of malaria, *Anopheles funestus*, *Anopheles gambiae* and *Anopheles coluzzii*, false positivity is less likely to arise because all of these species are anthropophilic. When alternative hosts are not available, the other important African malaria vector, *Anopheles arabiensis*, is also anthropophilic (Charlwood *et al.*, 1995). Furthermore, different laboratories may differ in their ELISA-based estimates of sporozoite rates (Kasasa *et al.*, 2013).

It is generally considered that sporozoite rates determined by ELISA can be estimated more accurately using the heads and thoraces of collected mosquitoes, rather than whole bodies. This is because abdomens may have mature sporozoites still inside oocysts rather than in the salivary glands (Verhave *et al.*, 1988). The separation of abdomen from head and thorax, however, takes a considerable amount of time and may not be necessary for comparative purposes. Similarly, in a study dealing with large samples of mosquitoes, running insects in pools of 10 specimens and correcting for any possible underestimation of rates saves considerable effort. It is not certain, however, that pool size does not affect estimated rates. Visual assessment of positive wells has previously been found to be as satisfactory as using a plate reader (Beier & Koros, 1991; Bockarie *et al.*, 1993), but it is not known whether naïve assessors can accurately identify positive mosquitoes.

In this study, a series of experiments to compare the effectiveness of the ELISA were performed using *An. funestus* and *An. gambiae s.l.* collected over a 10-year period from a village in southern Mozambique in order to determine the most cost-effective method of analysing large numbers of mosquitoes. The best strategy for sampling these mosquitoes based on estimated rates was also examined.

Materials and methods

Description of study site

The village of Furvela (24°43′ S, 35°18′ E), in Mozambique, *c.* 475 km north of the capital Maputo, has been described elsewhere (Charlwood *et al.*, 2003; Charlwood, 2011; Kampango *et al.*, 2011, 2013; Charlwood & Bragança, 2012). Most houses in the village are made of local materials (reeds and palm leaves) and rarely provide any barrier to entry by endophilic mosquitoes. Houses are either individually located or are built within family compounds of three or four houses. The village is bordered on two sides by extensive low floodplain, where mosquitoes breed. There was no electricity supply to the village at the time of the study. *Anopheles funestus* is the major vector in the village (Charlwood & Tomás, 2011), although in the first years of the study *An. gambiae s.l.* was also present (Charlwood, 2011).

Mosquito collection

From 2001 to 2010, host-seeking mosquitoes were collected using Centers for Disease Control (CDC) light traps hung approximately 1.5 m from the floor inside bedrooms close to the feet of beds in which people slept under mosquito nets. Householders were responsible for switching the traps on in the evening and off in the morning. Most trap nights occurred in the earlier years of the study.

From 2003 onwards, mosquitoes were also collected as they exited houses at dusk. The door of the house was left open and covered with a white mosquito-netting curtain and mosquitoes, which are positively phototropic at these times, were manually aspirated as they attempted to leave (Charlwood, 2011; Charlwood & Braganca, 2012). Different sets of 10–12 houses were sampled for approximately 4 days every week.

Collected anophelines were separated into species or species group according to the keys of Gillies & De Mellion (1968) and Gillies & Coetzee (1987), and sexed and counted. Females were further subdivided into ‘unfed’, ‘part-fed’, ‘fed’, ‘semi-gravid’ or ‘gravid’ categories according to the appearance of their abdomen as described by Detinova (1962).

Samples of *An. gambiae s.l.* obtained during 2002–2004 and 2008 were identified to species by PCR using the protocols of Scott *et al.* (1993) or Fanello *et al.* (2002). A small number of *An. funestus s.l.* was also identified to species using the protocol of Koekemoer *et al.* (2002).

Sporozoite determination

Unfed mosquitoes from the light trap collections and gravid females from the exit collections were stored according to species, date of collection, house number and collection method, in 1.5-mL microtubes over silica gel and subsequently analysed for the presence of *Plasmodium falciparum* circumsporozoite protein (CSP) by ELISA using protocols adapted from Wirtz *et al.* (1987).

Following overnight incubation at room temperature, 96-well U-bottomed microtitre plates coated with 50 μL solution of *P. falciparum* monoclonal antibodies (MAb) were drained, washed three times with phosphate-buffered saline (PBS) and Tween 20, filled with 200 μL blocking buffer (BB) and incubated for 1 h at room temperature. Mosquitoes were separated into groups of 10 or five, or single specimens, and transferred to new 1.5-mL microtubes. A quantity of 200 μL of boiled casein containing Igepal CA-630 was added to each microtube and the mosquitoes were ground by hand using a burnished glass rod. Plates were left in a freezer overnight, after which they were thawed and shaken before 50 μL of elutant was removed for the ELISA. Three positive and two negative control wells were included in each ELISA plate. Positive controls consisted of logarithmic increases in sporozoites (kindly supplied by Dr R. Wirtz) and negative controls consisted of PBS only and PBS plus the other reagents without mosquito elutant. Plates were incubated for 2 h and washed with PBS–Tween 20 twice. Aliquots of 50 μL of homologous peroxidase-conjugated MAb (0.05 μg/50 μL BB) were added to each well and incubated for 1 h. Plates were washed three times with PBS–Tween 20 and 100 μL ABTS peroxidase substrate was added to each well. All mosquitoes were analysed by the same technician.

Mosquitoes were generally analysed whole, in pools of 10 per reaction well. The samples therefore included specimens in which mature oocysts might be present but in which sporozoites would not be found in the salivary glands. In order to determine the extent of this overestimation, a subsample of mosquitoes from the collections was dissected and their heads and thoraces analysed (also in pools of 10) separately from their corresponding abdomens as per the method of Beier *et al.* (1987). Additionally, to determine if pool size had any effect on estimates, when individual collections had sufficient numbers for valid comparisons, individual and pools of five and 10 mosquitoes were analysed and compared. Other pool sizes were used when there were more (and occasionally fewer) than five mosquitoes remaining from a collection from which groups of 10 mosquitoes had been removed, or when fewer than 10 individuals were collected.

In 2006, plates were scanned with an ELISA plate reader (Model 550; BioRad Laboratories, Inc., Hercules, CA, U.S.A.) at 404–414 nm and read every 5 min until 30 min after their introduction into the reader (i.e. 60 min after the addition of the last reagent). Wells were considered positive if readings were more than three standard deviations (SDs) greater than the mean of the negative controls. In addition to the absorbance values, matrix reports were also retrieved from the ELISA plate readings. These consist of qualitative reports of the relative magnitude of the absorbances on the plate divided into 10 equal partitions, designated as grades 0–9.

Plates were additionally classified into four grades [negative (grade 0), and grades 1, 2 and 3] by two individuals working in a blinded fashion. They individually scored each plate immediately prior to its scanning by the plate reader and subsequently compared their readings. In cases of disagreement, the plate was jointly re-examined and a consensus reached. Only then was the result from the ELISA plate reader examined. Figure S1 shows an example of an individual plate. The ELISA plates for the samples collected and processed after 2006 were analysed only visually.

Underestimation of sporozoite rate according to pool size

As the overall sporozoite rate increases, so does the extent to which positivity rates are underestimated when mosquitoes are analysed in pools of 10. The extent of this underestimation can be determined according to Table S1, kindly provided by Professor Tom Smith of the Swiss Tropical and Public Health Institute, which was used to correct crude estimates of sporozoite rates.

'Adjusted' Wald intervals and 95% confidence intervals were obtained using the confidence interval calculator provided by Sauro (2005). The 'adjusted' Wald interval provides the best coverage for a given confidence interval when sample sizes are less than 150. It uses Wald's formula but is 'adjusted' in that it adds half of the squared Z-critical value to the numerator and the entire squared critical value to the denominator before computing the interval (Sauro & Lewis, 2005).

Ethics

The study was conducted under the aegis of the joint Instituto Nacional de Saúde (INS)–DBL Centre for Health Research and Development project 'Turning houses into traps for mosquitoes', which obtained ethical clearance from the National Bioethics Committee of Mozambique on 2 April 2001 (ref: 056/CNBS/01). Householders were informed about the purpose of the collections. Consent was obtained when collections were initiated.

Results

A total of 342 645 female *An. funestus* were collected in light traps and 200 170 were amassed from exit collections during the study. Numbers of *An. gambiae s.l.* collected amounted to 41 064 in light traps and 34 591 from exit collections. Other anopheline species collected in light traps included 5776 *Anopheles tenebrosus*, 725 *Anopheles letabensis*, 22 *Anopheles rufipes*, five *Anopheles squamosus*, and a single *Anopheles pharoensis*. A further 219 *An. tenebrosus* and five *An. rufipes* were also collected exiting houses.

Of the 437 *An. gambiae s.l.* identified to species during 2002–2004, 357 (82%) were *An. gambiae* [formerly S-form (Coetzee *et al.*, 2013)], 71 were *An. arabiensis* and nine were *Anopheles merus*. The mean ± SD proportion of

Table 1. Comparison between enzyme-linked immunosorbent assay (ELISA) plate reader matrix scores of circumsporozoite protein (cs)-ELISA wells and visually assessed scores of positivity in *Anopheles funestus* and *Anopheles gambiae s.l.* from Furvela, Mozambique.

	Visual reading, *n*				
Matrix score	Negative	Grade 1	Grade 2	Grade 3	Total, *n*
0	771	14	0	0	785
1	44	35	1	0	80
2	3	27	6	1	37
3	0	6	15	1	22
4	0	1	24	1	26
5	0	0	9	9	18
6	0	0	4	24	28
7	0	0	2	22	24
8	0	0	0	30	30
9	0	0	0	102	102
Total	818	83	61	190	1152

An. gambiae per month was 0.76 ± 0.12. Thus, all collections of *An. gambiae s.l.* probably included a mixture of *An. gambiae* and *An. arabiensis*. All of the *An. funestus* group identified morphologically were *An. funestus s.s.*, as were all 87 identified by PCR (A. L. Szlanski *et al.*, unpublished data, 2007). Therefore, *An. funestus s.s.* is considered to be the only member of this species group present in our collections.

A total of 41 792 of the *An. funestus* collected in light traps and 22 323 gravid females sourced from exit collections were analysed for CSP in the ELISA. In addition, 9431 *An. gambiae s.l.* collected in light traps and 6860 sourced from exit collections were also analysed.

Comparison between ELISA plate reader and visual readings

The concordance in the readings of positivity of the two assessors was almost complete. Differences between observers occurred over the classification of less than one well per plate. These mostly involved the labelling of a particular well as both 'negative' and 'grade 1 positive', or as 'grade 1 positive' and 'grade 2 positive'. Overall, the visual estimation of grades of intensity was consistent with that determined by the plate reader. Visual grade 1 corresponded to matrix grades 1–3, grade 2 to matrix grades 4–6, and grade 3 to matrix grades 7–9 (Table 1). Assuming that matrix grades 1 and above were all sporozoite-positive, sensitivity was 88.5%, specificity was 98.2% and accuracy was 94.6%. For ease of recording and analysis, only visual grades were used in the subsequent analyses.

Comparison of whole-body, head + thorax and abdomen-only analyses

Thirty-five (73%) of 48 samples of head + thorax positive pools were also positive for the matched abdomens. Samples from eight abdomens (14% of the paired positive samples) were positive only in abdomen analyses. This represents an overestimation of positivity in the overall whole-body samples of 0.6% (14% of a mean of 4.4%).

The results of the comparison of sporozoite rates and intensity of infection among whole-body, head + thorax and abdomen-only analyses are shown in Table 2. In both *An. funestus* and *An. gambiae*, the differences in the estimated intensity of infection between whole-body samples and head + thorax samples were not significant [Yates' corrected chi-squared test for whole-body vs. head + thorax: *An. funestus*, $\chi^2 = 1.82$ ($P = 0.177$); *An. gambiae s.l.*, $\chi^2 = 0.05$ ($P = 0.82$)]. However, in *An. funestus*, infection rates in either of these samples were significantly higher than that in abdomens alone [Yates' corrected chi-squared test: whole-body vs. abdomen-only samples, $\chi^2 = 15.13$ ($P = 0.0001$); head + thorax vs. abdomen-only samples, $\chi^2 = 17.51$ ($P < 0.0001$)]. For *An. gambiae s.l.*, however, abdomen-only samples were just as intensely infected as both whole-body samples and head + thorax samples [Yates' corrected chi-squared test: whole-body vs. abdomen-only samples, $\chi^2 = 0.77$ ($P = 0.380$); head + thorax vs. abdomen-only samples, $\chi^2 = 0.08$ ($P = 0.777$)]. This lack of difference was maintained even when grade 3 infections alone, which were the least common in abdominal samples, were considered (Fisher's two-tailed exact test: head + thorax vs. abdomen grade 3 infections, $P = 0.333$; whole-body vs. abdomen grade 3 infections, $P = 0.176$).

Rates for different pool sizes

Individually tested *An. funestus* had a significantly higher positivity rate than pool sizes of five or more mosquitoes [Yates' correction, $\chi^2 = 17.3$ ($P = 0.001$)], but *An. gambiae s.l.* did not [Yates' correction, $\chi^2 = 3.164$ ($P = 0.075$)]. Rates in the different pool sizes were not significantly different (Table 3).

Rates by species and collection method

Estimated rates of infection in both *An. funestus* and *An. gambiae s.l.* were higher in mosquitoes collected from light traps than in those obtained from exit collections over the same period (Table 4), but in neither case were these differences statistically significant [Yates' corrected chi-squared test for light trap vs. exit collections: *An. funestus*, $\chi^2 = 0.23$ ($P = 0.632$); *An. gambiae s.l.*, $\chi^2 = 0.45$ ($P = 0.502$)].

Although overall infection rates in *An. funestus* were higher than in *An. gambiae s.l.*, the differences between the two species, according to collection method, were not significant (Table 4) [Yates' corrected chi-squared test for *An. funestus* vs. *An. gambiae s.l.*: exit collections, $\chi^2 = 3.52$ ($P = 0.0606$); light trap collections, $\chi^2 = 2.4$ ($P = 0.12$)].

Discussion

Visual assessments of the positivity rate and grade of infection in the ELISA gave results comparable with those obtained

Table 2. Circumsporozoite protein enzyme-linked immunosorbent assay (cs-ELISA) results by visual grade, species and body part of mosquitoes and best estimate of positivity rate and 95% adjusted Wald confidence intervals (CIs) from light trap collections, including matched pairs, of head + thorax and abdomen-only investigations in *Anopheles funestus* and *Anopheles gambiae s.l.* from Furvela, Mozambique during 2004–2007.

Species		Negative	Grade 1	Grade 2	Grade 3 (proportion of total positive)	Total positive, *n*	Positive, %	95% CI
An. funestus	Whole mosquito	13 535	118	108	391 (0.63)	617	4.57%	4.22–4.92
	Head + thorax	1100	10	13	38 (0.63)	61	5.63%	4.33–7.07
	Abdomen	1123	10	9	3 (0.14)	22	2.04%	1.28–2.96
An. gambiae s.l.	Whole mosquito	5998	95	39	115 (0.46)	249	4.17%	4.22–4.92
	Head + thorax	247	0	2	7 (0.78)	9	4.02%	1.83–6.88
	Abdomen	252	4	1	2 (0.29)	7	3.15%	1.24–5.73

Table 3. Estimated sporozoite rates of *Anopeles funestus* and *Anopheles gambiae s.l.* from Furvela, Mozambique, by pool size. Corrected rates are based on the corrections for pools containing more than one infected mosquito based on Table S1 and associated equations (supporting information, online).

Species	Pool size	Tested, *n*	Positive, *n*	Best estimated raw rate, % (corrected rate %)	Adjusted 95% CI (corrected CI)
An. funestus	1	459	40	8.89%	6.40–11.67
	5	2195	105	4.78% (5.33%)	3.93–5.76 (4.46–6.35)
	8	448	14	3.33%	1.82–5.22
	9	621	26	4.33%	2.85–6.09
	10	9660	410	4.43% (4.74%)	3.85–4.65 (4.33–5.18)
An. gambiae s.l.	1	169	12	7.60%	4.00–12.11
	5	990	46	4.74% (5.26%)	3.49–6.15 (4.02–6.83)
	8	384	14	3.89%	2.13–6.08
	9	324	12	3.99%	2.06–6.43
	10	3990	155	3.91% (4.42%)	3.33–4.53 (3.82–5.09)

95% CI, 95% confidence interval.

from absorbance values measured with the ELISA plate reader. Similar results were obtained in previous studies conducted to determine positivity rates (Beier & Koros, 1991; Bockarie *et al.*, 1993). The present study also demonstrated that the classification grade used in visual assessment matched the classification of the grade of intensity given by the ELISA plate reader. This supports the suggestion that visual assessment can also give information on the intensity of infection, rather than simply a negative/positive result. Visual assessment has practical advantages over spectrophotometric analysis. It does not require the purchase and maintenance of sophisticated equipment and it allows for a quicker direct examination of results. The establishment of visual reading protocols that may represent more cost-effective measures in resource-poor national malaria control programmes has also been highlighted previously (Beier & Koros, 1991; Bockarie *et al*., 1993).

The over-representation of ELISA-positive mosquitoes in analyses using whole insects (eight of 48 pools) was similar to that found by Beier *et al.* (1987) (28 of 95 pools; $\chi^2 = 1$, d.f. = 1.23, $P = 0.2674$), which indicates that this is likely to be a relatively constant proportion. However, as found by Czeher *et al.* (2006), this proportion may vary by season. In the present study it resulted in a 0.6% overestimate of positivity as a result of the presence of mature sporozoites in oocysts that had yet to rupture. Although this might result in an underestimation of the absolute impact of a given anti-vector intervention, the use of whole mosquitoes in the ELISA saves considerable amounts of processing time. This means that more mosquitoes can be processed for the same amount of effort. In addition, the impact of this underestimation will be less important when the objective is to compare relative rates. Nevertheless, although the use of whole mosquitoes can be recommended for any large-scale study, it is recommended that baseline data on the underestimation rate is produced for the key time periods during which epidemiological monitoring occurs. In practice this means conducting ELISA tests on a random subsample of heads + thoraces from different seasons or locations. In addition, re-testing of positive samples by PCR may help to eliminate false positive samples obtained in the ELISA. This becomes especially important when the species concerned are only rarely incriminated in transmission, such as *Anopheles coustani* (Mwangangi *et al.*, 2013), *An. pharoensis* (Animut *et al.*, 2013) and *Anopheles melas* (Okorie *et al.*, 2011), for which sporozoite rates comparable with those of *An. gambiae* have been occasionally reported.

Differences between individually tested *An. gambiae* and those tested in pools of 10 were not significant. However, the

Table 4. Numbers of *Anopheles funestus* and *Anopheles gambiae s.l.* tested according to collection method and estimated sporozoite rates.

Species	Collection type	Tested, *n*	Positive, *n*	Rate, %	95% CI
An. funestus	Light trap	1701	52	3.11%	2.33–3.99
	Exit collection	6442	180	2.81%	2.42–3.23
An. gambiae s.l.	Light trap	2440	54	2.25%	1.70–2.88
	Exit collection	1319	24	1.89%	1.21–2.71

95% CI, 95% confidence interval.

significantly higher sporozoite rates observed in individually tested *An. funestus* in comparison with those tested in pools (i.e. adjusted rates) constituted an unexpected result. In the study village, there was no apparent effect of seasonality or year of collection that might explain this difference (data not shown). It may, therefore, reflect some sort of technical limitation of the study. It is possible that the mosquito grinding system used during the initial preparation of the biological homogenates may have been less efficient with smaller mosquitoes. A comparison of rates for different pool sizes using different grinding systems would seem apposite.

The results of this study support those of previous analyses showing that the use of individually tested head + thorax mosquito samples provides the most accurate estimates of sporozoite rate estimates (and hence EIR) (Beier *et al.*, 1987). However, even when the ELISA is 100% accurate, there is always a degree of uncertainty in the translation of these results into an estimate of EIR. Many factors can affect the likelihood that a mosquito with sporozoites in the salivary glands will transmit malaria. Some infected mosquitoes may not inject sporozoites when feeding (Beier & Koros, 1991) and even heavily infected mosquitoes may not inject a sufficient number of sporozoites to infect the vertebrate host (McElroy *et al.*, 1994). Thus an infected mosquito may not actually transmit the disease. Nevertheless, as with biting rates, relative values that refer to either time-points or regions (e.g. for control vs. intervention outcomes) rather than absolute values are usually required for monitoring purposes. Therefore, from the operational point of view, the shorter processing time and lower cost in terms of reagents and consumables per specimen may compensate for the loss of accuracy when either whole mosquitoes or mosquito pools are used.

The present results indicate that, for large-scale surveys, performing the ELISA using whole mosquitoes in pools of 10 and correcting for overestimation, with results noted according to a simple three-stage visual assessment of positivity, is the most cost-effective method of determining sporozoite rates in large samples. This process is sufficient to obtain comparative and reliable data. This is especially true in situations of *anophelism sans malaria*, or almost *sans malaria*, in which transmission is decreasing but mosquito numbers remain high, which is one possible scenario on the road to the elimination of malaria.

Supporting Information

Additional Supporting Information may be found in the online version of this article under the DOI reference: DOI: 10.1111/mve.12084

Figure S1. Photograph of an ELISA plate with the associated visual result.

Table S1. Underestimates of the proportion of mosquitoes positive for sporozoites in the ELISA when samples are grouped into pools of 10 (kindly provided by Professor Tom Smith of the Swiss Tropical and Public Health Institute).

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